THE MASTERMIND COACHING CLASSES

# Revision Notes

## Class 10 · Science

Physics and Environment — Chapters 9 to 13 · NCERT Science — Class 10 · CBSE / RBSE Syllabus 2026-27

**HOW TO USE THESE NOTES**

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

| CH. | CHAPTER TITLE | CORE THEME |
|---|---|---|
| **9** | Light: Reflection and Refraction | Mirrors, lenses, mirror and lens formulae, magnification, power |
| **10** | The Human Eye and the Colourful World | Eye, defects and correction, prism, dispersion, scattering |
| **11** | Electricity | Current, potential difference, Ohm's law, resistance, series and parallel, heating effect, power |
| **12** | Magnetic Effects of Electric Current | Field lines, field due to current, force on a conductor, motor, induction, generator, household circuits |
| **13** | Our Environment | Ecosystem, food chains, energy flow, ozone depletion, waste |

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CHAPTER 09 · PHYSICS

# Light: Reflection and Refraction

*We see the world because light reaches our eyes, and we can bend and direct it with mirrors and lenses. Torches, vehicle headlights, shaving mirrors, magnifying glasses, spectacles and cameras all depend on these ideas. This chapter covers the laws of reflection and refraction, the images formed by spherical mirrors and lenses, the sign convention, the formulae for image position and size, and the power of a lens.*

## Big Questions

- How do spherical mirrors form images and how do we find them?
- Why does light bend when it changes medium?
- How do lenses form images and how is their power defined?

## 9.1 Reflection of Light

DEFINITION

**Laws of reflection:** (i) the angle of incidence equals the angle of reflection; (ii) the incident ray, the reflected ray and the normal at the point of incidence lie in the **same plane**.

A plane mirror forms an image that is **virtual, erect, laterally inverted**, the same size as the object and as far behind the mirror as the object is in front.

## 9.2 Spherical Mirrors

| Term | Meaning |
|---|---|
| **Pole (P)** | Centre of the mirror surface |
| **Centre of curvature (C)** | Centre of the sphere of which the mirror is a part |
| **Radius of curvature (R)** | Radius of that sphere; **R = 2f** |
| **Principal axis** | Line through P and C |
| **Principal focus (F)** | Point where rays parallel to the axis meet (concave) or appear to come from (convex) |
| **Focal length (f)** | Distance PF |

FORMULA

**f = R/2. Mirror formula:** $1/v + 1/u = 1/f$. **Magnification:** $m = h'/h = -v/u$.

**Ray 1**
A ray parallel to the principal axis passes through F (concave) or appears to come from F (convex) after reflection.

**Ray 2**
A ray through F (concave) emerges parallel to the axis.

**Ray 3**
A ray through C returns along its own path.

**Ray 4**
A ray towards P is reflected making equal angles with the axis.

### ▸ Sign convention (New Cartesian)

- Pole is the origin; the principal axis is the x-axis; the object is placed on the left so that **u is negative**.
- Distances to the right of P are positive, to the left negative.
- Heights above the axis are **positive**, below are negative.
- For a concave mirror **f is negative**, for a convex mirror **f is positive**.

| Object position (concave mirror) | Image position | Nature and size |
|---|---|---|
| **At infinity** | At F | Real, inverted, highly diminished |
| **Beyond C** | Between F and C | Real, inverted, diminished |
| **At C** | At C | Real, inverted, same size |
| **Between C and F** | Beyond C | Real, inverted, enlarged |
| **At F** | At infinity | Real, inverted, highly enlarged |
| **Between P and F** | Behind the mirror | Virtual, erect, enlarged |

A **convex mirror** always gives a **virtual, erect, diminished** image between P and F. Uses: concave mirrors in torches, headlights, shaving mirrors, dentists' mirrors and solar furnaces; convex mirrors as **rear-view mirrors** because they give a wider field of view.

## 9.3 Refraction of Light

When light travels obliquely from one transparent medium to another it changes direction because its speed changes. This is **refraction**. Going from rarer to denser medium (air to glass) the ray bends **towards the normal**; denser to rarer, **away from the normal**.

FORMULA

**Snell's law:** sin i / sin r = constant (**refractive index n**) for a pair of media. **Absolute refractive index:** n = c/v (speed of light in vacuum divided by speed in the medium). Air n = 1.0, water 1.33, glass about 1.5, diamond 2.42. Light travels slowest in the medium with the highest n (optically densest).

- **Rectangular glass slab:** emergent ray is parallel to the incident ray, displaced sideways (lateral displacement).
- A coin in a water tank looks raised; a pencil dipped in water appears bent, because of refraction.

## 9.4 Spherical Lenses

A **convex lens** (converging) is thicker at the centre; a **concave lens** (diverging) is thinner at the centre. Each has an **optical centre (O)**, two **principal foci** ($F_1$, $F_2$) and a focal length f.

**Ray 1**
A ray parallel to the axis passes through F (convex) or appears to come from F (concave).

**Ray 2**
A ray through the optical centre goes straight.

**Ray 3**
A ray through F emerges parallel to the axis.

| Object position (convex lens) | Image position | Nature and size |
|---|---|---|
| **At infinity** | At $F_2$ | Real, inverted, highly diminished |
| **Beyond $2F_1$** | Between $F_2$ and $2F_2$ | Real, inverted, diminished |
| **At $2F_1$** | At $2F_2$ | Real, inverted, same size |
| **Between $F_1$ and $2F_1$** | Beyond $2F_2$ | Real, inverted, enlarged |
| **At $F_1$** | At infinity | Real, inverted, highly enlarged |
| **Between $F_1$ and O** | Same side as object | Virtual, erect, enlarged (magnifying glass) |

A **concave lens** always forms a virtual, erect, diminished image between O and $F_1$.

FORMULA

**Lens formula:** $1/v - 1/u = 1/f$. **Magnification:** $m = h'/h = v/u$. A **convex** lens has **positive** f; a **concave** lens **negative** f. **Power:** $P = 1/f$ (f in metres); unit is the **dioptre (D)**. Convex P is positive, concave P negative; for lenses in contact $\mathbf{P = P_1 + P_2}$.

## 9.5 Worked Examples

**Concave mirror.** An object 4 cm tall is 30 cm from a concave mirror of f = −15 cm. From $1/v = 1/f - 1/u = -1/15 + 1/30 = -1/30$, **v = −30 cm** (real image at the same place); $m = -v/u = -(-30)/(-30) = -1$, so the image is 4 cm tall, **inverted, the same size**.

**Convex lens.** Object at u = −20 cm, f = +10 cm. $1/v = 1/f + 1/u = 1/10 - 1/20 = 1/20$, so **v = +20 cm**; $m = v/u = -1$: real, inverted, same size (object at 2F).

**Power.** A lens of focal length −50 cm has $P = 1/(-0.5) =$ **−2 D**, a diverging lens. A lens of +2 D has f = 0.5 m = 50 cm. Two lenses of +2 D and −0.5 D in contact have P = **+1.5 D**.

**Refractive index.** If light travels at $2 \times 10^8$ m/s in a medium, $n = 3 \times 10^8 / 2 \times 10^8 =$ **1.5**.

## Quick Revision: Likely Exam Questions

| Likely question | Model answer |
|---|---|
| **State the mirror formula.** | $1/v + 1/u = 1/f$. |
| **Which mirror is used in vehicles as rear-view and why?** | Convex mirror: erect, diminished image with a wide field of view. |
| **Define one dioptre.** | Power of a lens whose focal length is 1 m. |
| **Where is the object placed to get a real image of the same size using a concave mirror?** | At the centre of curvature. |
| **What is the nature of the image by a concave lens?** | Virtual, erect, diminished. |
| **What is the speed of light in diamond?** | $c/2.42$ = about $1.24 \times 10^8$ m/s. |
| **Why does a pencil in water look bent?** | Light bends when it moves from water into air. |

| Likely question | Model answer |
|---|---|
| **Which lens is used as a magnifier?** | Convex lens with the object between F and O. |

### ★ KEY TAKEAWAYS

- Reflection: **angle i = angle r**; spherical mirror **f = R/2**; **1/v + 1/u = 1/f**; m = −v/u.
- Refraction: **bends towards the normal** in a denser medium; **n = c/v**; Snell's law sin i / sin r = constant.
- Lens: **1/v − 1/u = 1/f**; m = v/u; **power P = 1/f** in dioptres.
- Convex lens or concave mirror gives real or virtual images; convex mirror and concave lens always give **virtual, erect, diminished**.

CHAPTER 10 · PHYSICS

# The Human Eye and the Colourful World

*The eye is a natural camera whose lens forms an image on a light-sensitive screen. Spectacles correct its defects; prisms split sunlight into a rainbow, and scattering by air explains the blue sky, the red sun at sunset and the path of a torch beam in a dusty room. This chapter uses refraction to understand all of these.*

## Big Questions

- How does the human eye form an image and adjust its focus?
- What are the common defects of vision and how are they corrected?
- How do a prism, a rainbow and the scattering of light create colours?

## 10.1 The Human Eye

| Part | Function |
|---|---|
| **Cornea** | Transparent front surface; most of the refraction of light happens here |
| **Iris** | Coloured muscular diaphragm that controls the pupil |
| **Pupil** | Opening whose size changes with light intensity |
| **Eye lens** | Flexible convex lens; fine focusing; adjusts focal length |
| **Ciliary muscles** | Change the curvature of the lens |
| **Retina** | Light-sensitive screen of **rod cells** (dim light) and **cone cells** (colour) |
| **Optic nerve** | Carries electrical signals to the brain, which interprets the inverted image as upright |

The ability of the eye lens to change its focal length to see near and distant objects clearly is called **accommodation**. For a distant object the ciliary muscles relax, the lens becomes thin (long focal length). For a near object they contract, the lens becomes thicker (shorter focal length).

DEFINITION

**Near point:** the minimum distance for clear vision, about **25 cm** for a normal eye. **Far point:** the farthest point seen clearly, **infinity** for a normal eye. Persistence of vision is about 1/16 second.

## 10.2 Defects of Vision and Their Correction

| Defect | Cause | Image formed | Correction |
|---|---|---|---|
| **Myopia (short-sightedness)** | Eyeball too long, or lens too curved (too short focal length) | In **front of** retina; distant objects blurred | **Concave** lens of suitable power |
| **Hypermetropia (far-sightedness)** | Eyeball too short, or focal length too long | **Behind** retina; near objects blurred | **Convex** lens |
| **Presbyopia** | Weak ciliary muscles and lens stiffening with age | Both near and far difficulty | **Bifocal lenses** (concave upper, convex lower) |

- **Cataract:** the lens becomes milky and cloudy in old age, causing blurred vision; cured by surgery.
- **Eye donation:** the cornea of a deceased person can restore the sight of the corneal blind; donors may be of any age, and the eyes should be removed within 4 to 6 hours.
- In a myopic eye the far point is nearer than infinity, so a diverging lens makes the image of a distant object appear at that far point.

FORMULA

To correct myopia with far point at distance D, use a concave lens of **f = −D**; power P = −1/D. A person with far point 80 cm needs P = −1/0.8 = **−1.25 D**.

## 10.3 Refraction through a Prism and Dispersion

A glass **prism** has two triangular bases and three rectangular faces. A ray entering bends towards the normal and, on leaving, away from it, so it is deviated towards the **base**. The angle between the incident and emergent rays is the **angle of deviation**; the angle between the two refracting faces is the **angle of the prism**.

DEFINITION

**Dispersion** is the splitting of white light into its component colours. The band of seven colours is the **spectrum**: **VIBGYOR** (violet, indigo, blue, green, yellow, orange, red). **Red bends least** and **violet bends most** because different colours travel at different speeds in glass.

- Newton used a second inverted prism to recombine the colours into white light, proving that the prism does not colour light but only splits it.
- A **rainbow** is a natural spectrum seen after rain when the Sun is behind the observer: raindrops act as small prisms that disperse (refract), internally reflect and refract the light again. Red is on the outside and violet inside.

## 10.4 Scattering of Light

Light is scattered when it strikes particles in a medium. **Tyndall effect:** the beam of light becomes visible due to scattering by fine particles (dust, smoke, water droplets) in a colloid; seen in a dusty room, sunlight through a forest canopy.

| Phenomenon | Explanation |
|---|---|
| **Blue colour of the sky** | Air molecules are smaller than the wavelength of visible light and scatter **shorter wavelengths (blue)** much more than red |
| **Red colour of Sun at sunrise and sunset** | Sun is near the horizon, light passes a longer path in the atmosphere, most blue is scattered away and red reaches us |
| **White clouds** | Droplets are large and scatter all colours alike |
| **Sky is dark for astronauts** | No atmosphere in space to scatter |
| **Danger signals are red** | Red is scattered least by fog and smoke, so it is seen from far |

IMPORTANT

Scattering is inversely related to wavelength: blue (about 450 nm) is scattered more than red (about 700 nm).

## 10.5 Worked Examples

**Spectacle power.** A hypermetropic eye has near point 50 cm and the normal near point is 25 cm. The lens must form the image of an object at 25 cm at 50 cm: u = −25, v = −50, 1/f = 1/v − 1/u = −1/50 + 1/25 = 1/50, **f = +50 cm, P = +2 D** (convex).

**Why do we see only red in a sunset?** More blue is scattered on the long path, so the transmitted light is rich in red.

**Why can't we see the rainbow at noon?** Sun must be behind us at a low altitude (below about 42 degrees) for the rays to reach us.

## Quick Revision: Likely Exam Questions

| Likely question | Model answer |
|---|---|
| **What is accommodation?** | The eye lens changing its focal length to focus near and distant objects. |
| **Why does a myopic eye need a concave lens?** | The image forms in front of the retina; a diverging lens pushes it back onto the retina. |
| **Which colour bends most in a prism?** | Violet. |
| **Why is the sky blue?** | Blue light of short wavelength is scattered most by air molecules. |
| **What is Tyndall effect?** | Scattering of light by colloidal particles that makes the path visible. |
| **Which cells detect colour?** | Cone cells. |
| **What is the least distance of distinct vision?** | About 25 cm. |
| **How is cataract cured?** | By surgical replacement of the cloudy lens. |

★ KEY TAKEAWAYS

- Eye: **cornea refracts, lens fine-tunes, retina forms the image**; the lens changes focal length by **accommodation**.
- **Myopia → concave lens; hypermetropia → convex lens; presbyopia → bifocal.**
- Prism disperses white light into **VIBGYOR**; red bends least, violet most.
- **Scattering** explains the blue sky, red sunrise and sunset, and the Tyndall effect.

CHAPTER 11 · PHYSICS

# Electricity

*Lamps, fans, heaters and mobile chargers all run on the flow of electric charge. Understanding current, voltage and resistance lets us design circuits, choose the right wire and fuse, and calculate the electricity bill. This chapter builds the basic quantities, states Ohm's law, compares series and parallel circuits and uses the heating effect of current.*

## Big Questions

- What is electric current and what drives it?
- How do resistances combine in series and parallel?
- How much heat and energy does a current produce?

## 11.1 Electric Current and Potential Difference

DEFINITION

**Electric current** is the rate of flow of charge: **I = Q/t**. One **ampere** is one coulomb of charge flowing per second. The charge on one electron is 1.6 x $10^{-19}$ C, so 1 C is about 6 x $10^{18}$ electrons. Conventional current flows from the positive terminal to the negative, opposite to the flow of electrons.

DEFINITION

**Potential difference** between two points is the work done to move one coulomb of charge between them: **V = W/Q**. One **volt** is one joule per coulomb. A **cell** or battery maintains the potential difference; it is measured by a **voltmeter** connected in **parallel**, while current is measured by an **ammeter** connected in **series**.

| Component | Symbol idea | Role |
|---|---|---|
| **Cell / battery** | Long line (+), short line (−) | Source of potential difference |
| **Plug key (switch)** | Open or closed | Completes or breaks the circuit |
| **Ammeter** | A in a circle | Measures current; low resistance; in series |
| **Voltmeter** | V in a circle | Measures potential difference; high resistance; in parallel |
| **Resistor** | Zig-zag or box | Opposes current |
| **Rheostat** | Variable resistor | Changes current |

## 11.2 Ohm's Law and Resistance

FORMULA

**Ohm's law:** at constant temperature, **V ∝ I**, i.e. **V = IR**. **R = V/I**; the SI unit of resistance is the **ohm (Ω)**; 1 Ω is the resistance when 1 V gives 1 A. The V–I graph of an ohmic conductor is a straight line through the origin; its slope is 1/R.

Resistance arises because moving electrons collide with the atoms of the conductor. It depends on four things.

FORMULA

**R = ρL/A**, where ρ is **resistivity** (unit Ω m). Resistance **increases** with length and with temperature (for metals) and **decreases** with cross-section area.

| Type | Resistivity range (Ω m) | Examples |
|---|---|---|
| **Conductors** | $10^{-8}$ to $10^{-6}$ | Silver, copper, aluminium |
| **Alloys** | Higher than their metals, hardly oxidise when hot | Nichrome, manganin, constantan |
| **Insulators** | $10^{12}$ to $10^{17}$ | Rubber, glass, ebonite |

- Alloys such as **nichrome** are used in heating elements because of high resistivity and no oxidation at high temperature.
- Copper and aluminium are used for **transmission lines** because of low resistivity.
- Tungsten has a high melting point (3380 °C), so it is used for **bulb filaments**.

## 11.3 Combination of Resistors

| Feature | Series | Parallel |
|---|---|---|
| **Current** | Same through each | Divides among branches |
| **Potential difference** | Divides across resistors | Same across each |
| **Equivalent resistance** | $\mathbf{R_s = R_1 + R_2 + R_3}$ | $\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3}$ |
| **Value** | Greater than the largest | Less than the smallest |
| **Failure** | One break stops all | Others continue |
| **Use** | Fuse with appliance | Home wiring |

IMPORTANT

Domestic circuits use **parallel** connection so each appliance gets the full 220 V, works independently and draws its own current; in series a break in one appliance would stop all.

## 11.4 Heating Effect and Electric Power

FORMULA

**Joule's law:** $H = I^2Rt$ (joules). **Power:** $P = VI = I^2R = V^2/R$, unit **watt (W)**. **Energy:** E = P x t; **1 kWh = 3.6 x $10^6$ J**, the commercial unit (one unit of electricity).

- Heating effect powers the electric heater, iron, toaster, geyser and the bulb filament.
- A **fuse** is a wire of low melting point and low rating in series; it melts and breaks the circuit during overloading or short-circuit and protects appliances.
- The same **current** flows in series, so heat is greatest in the **largest** resistor ($H \propto R$); in parallel, with the same voltage, heat is greatest in the **smallest** resistor ($H = V^2t/R$).

## 11.5 Worked Examples

**Ohm's law.** A resistor of 20 Ω across a 12 V battery: I = 12/20 = **0.6 A**; power = VI = 12 x 0.6 = **7.2 W**.

**Series.** 2 Ω, 3 Ω and 5 Ω in series: R = **10 Ω**; with a 20 V supply I = 2 A and V across the 3 Ω resistor is 6 V.

**Parallel.** 6 Ω and 3 Ω: 1/R = 1/6 + 1/3 = 1/2, **R = 2 Ω**; with a 12 V supply the branch currents are 2 A and 4 A, total **6 A**.

**Resistivity.** A wire of resistivity 1.6 x $10^{-8}$ Ω m, length 10 m and area 1 x $10^{-6}$ $m^2$ has R = 1.6 x $10^{-8}$ x 10 / $10^{-6}$ = **0.16 Ω**. If the wire is stretched to double its length, volume is constant so the area halves and **R becomes four times**.

**Energy bill.** A 1000 W heater used 2 h a day for 30 days: energy = 1 x 2 x 30 = 60 kWh; at a rate of &rupee;6 per unit, cost = **&rupee;360**.

**Heat produced.** A 100 Ω resistor carrying 2 A for 5 minutes: H = $I^2Rt$ = 4 x 100 x 300 = **1.2 x $10^5$ J**.

## Quick Revision: Likely Exam Questions

| Likely question | Model answer |
|---|---|
| **Define potential difference.** | Work done per unit charge in moving a charge between two points: V = W/Q. |
| **Why is the voltmeter connected in parallel?** | It must read the potential difference across a component and have very high resistance. |
| **Why is tungsten used for filaments?** | High melting point and high resistivity. |
| **State Joule's law.** | H = $I^2Rt$. |
| **How many joules in 1 kWh?** | 3.6 x $10^6$ J. |
| **Why do we connect household appliances in parallel?** | Equal voltage, independent working. |
| **What is the function of a fuse?** | It melts and breaks the circuit when current exceeds a safe value. |
| **What happens to resistance if length is halved?** | It becomes half (R ∝ L). |

★ KEY TAKEAWAYS

- **I = Q/t; V = W/Q; V = IR.**
- **R = ρL/A**; alloys have higher resistivity than pure metals.
- Series: **$R = R_1 + R_2$**; parallel: **$1/R = 1/R_1 + 1/R_2$**.
- **H = $I^2Rt$; P = VI; 1 kWh = 3.6 x $10^6$ J.**
- Ammeter in **series**, voltmeter in **parallel**; household wiring is parallel with a fuse.

CHAPTER 12 · PHYSICS

# Magnetic Effects of Electric Current

*A compass needle swings when a current-carrying wire is nearby: electricity and magnetism are linked. This link powers electric motors in fans and mixers, generators in power stations, transformers on poles and the circuits inside our homes. The chapter studies magnetic fields, the field produced by currents, the force on a current in a field, and electromagnetic induction.*

## Big Questions

- How does an electric current produce a magnetic field?
- How does a magnetic field exert force and run a motor?
- How is electric current produced by changing magnetic fields?

## 12.1 Magnetic Field and Field Lines

A magnet has two poles, north-seeking and south-seeking; like poles repel, unlike poles attract. The region around a magnet in which its force can be felt is the **magnetic field**. Field lines are traced with iron filings or a compass.

- Field lines emerge from the **north pole** and merge at the **south pole** outside the magnet; inside they go from south to north, forming **closed curves**.
- The **closeness** of lines shows the strength of the field; field lines never **intersect**, since the compass can point in only one direction at a place.

## 12.2 Magnetic Field due to a Current

**Oersted (1820)** found that a compass needle near a wire deflects when current flows, and the deflection reverses when the current reverses.

DEFINITION

**Right-hand thumb rule:** hold the straight conductor in the right hand with the thumb along the current; the fingers curl in the direction of the **magnetic field** lines.

| Conductor | Field pattern | Dependence |
|---|---|---|
| **Straight wire** | Concentric circles around the wire | Stronger with larger current and nearer the wire |
| **Circular loop** | Circles near the wire become larger and larger farther away; straight at the centre | Stronger with more turns, larger current, smaller radius |
| **Solenoid (long coil)** | Like a bar magnet; **uniform** field inside, parallel straight lines | Stronger with more turns and larger current; soft iron core makes an **electromagnet** |

- A solenoid has a north and south pole like a bar magnet; one end behaves as north.
- A **permanent magnet** is made of steel, an **electromagnet** of soft iron, whose magnetism disappears when current stops.

## 12.3 Force on a Current-carrying Conductor

A current-carrying rod in a magnetic field experiences a force; **André-Marie Ampère** suggested that a magnet exerts an equal and opposite force on the conductor. The force is **maximum** when the current is at **right angles** to the field, and its direction reverses if the current or the field direction is reversed.

> DEFINITION
> **Fleming's left-hand rule:** stretch the **forefinger, middle finger and thumb** of the left hand at right angles. If the forefinger points in the direction of the **field** and the middle finger in the direction of the **current**, the thumb gives the direction of the **force** (motion).

### ▸ Electric motor

A motor converts **electrical energy to mechanical energy**. A rectangular coil (armature) is between the poles of a magnet; the two sides carry current in opposite directions, so they experience opposite forces and the coil rotates. A **split-ring commutator** reverses the current every half turn, so the coil keeps turning in the same direction. Brushes link the commutator with the battery. Motors drive fans, washing machines, mixers, pumps and many other devices; strength increases with a stronger magnet, more turns, more current and soft iron core.

## 12.4 Electromagnetic Induction

> DEFINITION
> **Electromagnetic induction** is the production of an induced potential difference (and current) in a coil when the magnetic field around it **changes** (Faraday, 1831). It is produced by moving a magnet relative to a coil, or by changing the current in a nearby coil.

- A galvanometer needle deflects only while the magnet is moving; deflection is opposite when the magnet is withdrawn.
- **Fleming's right-hand rule:** forefinger = field, thumb = motion of conductor, middle finger = direction of **induced current**.

### ▸ Electric generator

A generator converts **mechanical energy into electrical energy**. A coil rotating in a magnetic field cuts field lines, and an induced current flows; the current changes direction after every half rotation, giving **alternating current (AC)**. With **slip rings** the output is AC; with a **split-ring commutator** it is **direct current (DC)**.

| Feature | AC | DC |
|---|---|---|
| **Direction** | Reverses periodically; in India every 1/100 s (frequency **50 Hz**) | Fixed direction (cell, battery) |
| **Source** | Power stations, generators | Cells, batteries, solar cells |
| **Advantage** | **Transmitted over long distances** with little loss | Used in electronics |

## 12.5 Domestic Electric Circuits

- Supply from the mains reaches homes through a **live wire (red, positive)** and **neutral wire (black)** with a potential difference of 220 V; the **earth wire (green)** is connected to a metal plate deep in the earth.
- The **earth wire** gives a low-resistance path for leakage current to the ground, so the body of the appliance (kettle, refrigerator) stays at earth potential and the user avoids shock.

- **Short circuit:** live and neutral touch, current becomes very large; **overloading:** too many appliances on one socket, drawing excess current. A **fuse** or an MCB protects against both.
- Circuits for each room are **parallel**; a 15 A line is used for heavy appliances (geyser, AC), 5 A for bulbs and fans.

## 12.6 Worked Examples

**Direction of force.** Current flows from east to west in a wire placed in a vertical field pointing downward. The force is perpendicular to both, so it acts along the north-south line; applying the left-hand rule (F = I L x B) gives **south**.

**Choosing the rule.** Motor: current and field known, find force → **left-hand rule**. Generator: motion and field known, find induced current → **right-hand rule**.

**Why does a coil rotate continuously in a motor?** The commutator reverses the current every half turn, so the force on each side keeps the same turning sense.

## Quick Revision: Likely Exam Questions

| Likely question | Model answer |
|---|---|
| **What did Oersted observe?** | Current in a wire deflects a nearby compass needle, so currents produce magnetic fields. |
| **State Fleming's left-hand rule.** | Forefinger: field; middle finger: current; thumb: force. |
| **Why do magnetic field lines never cross?** | Two tangents at a point would give two field directions, which is impossible. |
| **What is the function of the commutator?** | Reverses the current every half rotation in the motor. |
| **Name two safety devices in a home circuit.** | Fuse and earth wire (also MCB). |
| **What is the frequency of AC in India?** | 50 Hz. |
| **What is the role of the soft iron core in an electromagnet?** | It becomes strongly magnetised and strengthens the field. |
| **What is the difference between AC and DC generators?** | AC uses slip rings; DC uses a split-ring commutator. |

★ KEY TAKEAWAYS

- Current produces a magnetic field; the **right-hand thumb rule** gives its direction.
- A **solenoid** carrying current behaves like a bar magnet; iron core gives an **electromagnet**.
- **Force on conductor:** Fleming's **left-hand rule**; the **motor** converts electrical to mechanical energy.
- **Induction:** changing field produces current; **right-hand rule**; generator converts mechanical to electrical energy.
- Home wiring: **live, neutral, earth**; fuse prevents overloading.

CHAPTER 13 · BIOLOGY

# Our Environment

*A pond, a forest, a village field and even an aquarium are small worlds in which living things and their surroundings interact. Human activity changes these worlds, sometimes harmfully. This chapter explains what an ecosystem is, how energy and matter flow through food chains, and why problems such as ozone depletion and waste accumulation need our attention.*

## Big Questions

- What are the components of an ecosystem?
- How do energy and chemicals move through food chains?
- How do human activities harm the environment, and what can we do?

## 13.1 What is an Ecosystem?

DEFINITION

An **ecosystem** is a community of interacting **living (biotic)** organisms together with the **non-living (abiotic)** factors such as temperature, rainfall, sunlight, wind and soil. Ecosystems can be natural (forest, pond, lake) or **artificial** (garden, crop field, aquarium).

| Role | Description | Examples |
|---|---|---|
| **Producers** | Make food from sunlight by photosynthesis; the first link | Green plants, algae |
| **Consumers** | Depend on others for food | Herbivores, carnivores, omnivores, parasites |
| **Decomposers** | Break down dead plants and animals into simple substances, returning nutrients to the soil | Bacteria, fungi |

- **Decomposers** are the cleaning agents of nature; without them dead matter would pile up and the soil would not be replenished.
- Consumers are **primary** (herbivores), **secondary** (feed on herbivores), **tertiary** (feed on secondary consumers).

## 13.2 Food Chains and Webs

A **food chain** is a sequence of organisms in which each eats the one before it. Each step is a **trophic level**: Grass → Grasshopper → Frog → Snake → Hawk. In a pond, Algae → Small fish → Large fish. Several interlinked chains form a **food web**.

| Trophic level | Organism | Type |
|---|---|---|
| **First** | Plants | Producer |
| **Second** | Herbivores | Primary consumer |
| **Third** | Small carnivores | Secondary consumer |
| **Fourth** | Large carnivores | Tertiary consumer |

## 13.3 Flow of Energy

The Sun is the source of energy. Producers capture only about **1 per cent** of the solar energy that falls on leaves. Energy flows in **one direction**: it never returns to the Sun.

> **FORMULA**
> **Ten per cent law:** only about **10 per cent** of the energy at one trophic level is passed on to the next; the rest is lost as heat in respiration and in other life activities. Starting with 10,000 J in producers: 1000 J (herbivores), 100 J (carnivores), 10 J (top carnivores).

- Because of this loss, food chains have only **3 to 4 trophic levels**; there is too little energy for a fifth level.
- **Pyramid of energy** is always upright, wide at the base and narrow at the top.
- Eating at lower trophic levels (grains and plants) provides more energy for the same land area.

### ▸ Biological magnification

Pesticides and other harmful chemicals washed from fields enter water and the soil, and are absorbed by plants. They are not broken down, so they **accumulate** at every level and are most concentrated in the **top consumers** (humans). This increase in concentration of non-degradable chemicals along the food chain is **biological magnification**. Cereals, pulses, fruits and milk have pesticide residues, so thorough washing and organic farming help.

## 13.4 Environmental Problems

### ▸ Ozone layer depletion

**Ozone ($O_3$)** is a molecule of three oxygen atoms. At the top of the atmosphere it forms from oxygen by high-energy ultraviolet (UV) radiation, and it **shields the Earth from harmful UV rays**. UV can cause skin cancer, cataract and damage to crops. The ozone layer got thinner because of **chlorofluorocarbons (CFCs)** used in refrigerators, air conditioners and aerosol sprays. In 1987 the UN Environment Programme agreed to freeze CFC production, and later to phase it out.

> **FORMULA**
> $O_2 \rightarrow O + O$ (UV) and $O_2 + O \rightarrow O_3$. Ozone absorbs UV. CFCs release chlorine atoms that destroy ozone.

### ▸ Waste

| Type | Meaning | Examples | Effect / handling |
|---|---|---|---|
| **Biodegradable** | Broken down by bacteria, fungi and other agents | Vegetable waste, paper, cotton, wool, wood, animal waste | Compost, manure; turns into harmless products |
| **Non-biodegradable** | Not broken down by natural processes | Plastics, polythene, glass, metals, DDT, aluminium cans, synthetic fibres | Stay for years, harm soil and water; choke animals; they accumulate |

- Waste is produced in large quantity by households, factories and hospitals. Use **dustbins**, separate wet and dry waste, **reduce, reuse and recycle**.
- Polythene bags choke drains and are eaten by cattle; avoid single-use plastic; use cloth or jute bags.
- Every action has an effect: **dump sites** pollute land; burning gives smoke.

## 13.5 Worked Examples

**Energy in a chain.** Producers hold 20,000 J. Primary consumers get 2000 J, secondary consumers **200 J**, tertiary consumers **20 J** (10 per cent law).

**Which chain is longer?** A chain with 5 steps is rare because the energy at the fifth level would be 0.01 per cent of the producers' energy.

**Why is DDT in human milk higher than in grass?** Biological magnification: humans are at the top of the chain so the poison concentrates.

## Quick Revision: Likely Exam Questions

| Likely question | Model answer |
|---|---|
| **Name the components of an ecosystem.** | Biotic (producers, consumers, decomposers) and abiotic. |
| **What is the ten per cent law?** | Only 10 per cent of energy goes to the next trophic level. |
| **Why are food chains short?** | Energy lost at each level leaves too little for more levels. |
| **What is biomagnification?** | Increase in concentration of harmful chemicals at successive trophic levels. |
| **Which gas protects us from UV?** | Ozone ($O_3$). |
| **Which chemicals deplete ozone?** | Chlorofluorocarbons (CFCs). |
| **Give two examples of non-biodegradable waste.** | Plastic, DDT, glass, aluminium cans. |
| **What do decomposers do?** | Break down dead matter and return nutrients to the soil. |

★ KEY TAKEAWAYS

- Ecosystem = **biotic + abiotic**; roles: **producers, consumers, decomposers**.
- Energy flows **one way**; only **10 per cent** moves to the next trophic level.
- **Biological magnification** concentrates non-degradable chemicals in top consumers.
- **CFCs** deplete the **ozone** layer that blocks UV rays.
- Reduce, reuse, recycle; separate **biodegradable** and **non-biodegradable** waste.